Sample Project Documentation

Phase 7: Project Documentation

Abstract

Trip Genius AI is a web-based travel planning application that uses a generative model to create a structured itinerary from a traveller’s preferences. The user provides the starting city, destination, dates, budget, traveller count, interests and travel style. The backend calls Groq, validates the generated structure and stores trip and itinerary records in Supabase. The frontend supports saved trips, printing and Google Maps searches for hotels and restaurants.

1. Introduction

Planning a trip involves balancing time, cost and interests. A collection of destination suggestions is not enough; the traveller also needs to distribute activities across days and keep a plan available for review. Trip Genius AI addresses the initial planning step by creating an organized itinerary draft from a single set of inputs.

2. Objectives

- Reduce the manual effort needed to create a first itinerary draft.
- Use trip preferences to generate a day-wise plan.
- Provide estimated costs and travel tips.
- Persist trips so they can be revisited.
- Offer external map searches for hotel and restaurant research.
- Allow the itinerary to be printed or saved as a PDF.

3. Technology and Architecture

The frontend uses React, TypeScript and Vite. The backend uses Python and FastAPI. Groq provides the generative model endpoint, configured as openai/gpt-oss-20b. Pydantic schemas define expected data structures, and Supabase stores trips and itineraries. The frontend is deployed on Vercel and the backend on Render.

4. Methodology

1. Collect the trip preferences through the frontend form.
2. Send the values to the FastAPI itinerary endpoint.
3. Construct a prompt and call the Groq model.
4. Parse the model output and validate required fields.
5. Save the trip and itinerary in Supabase.
6. Return the itinerary to the frontend for display.
7. Allow the user to save, reopen, delete, print and use external map searches.

5. Functional Modules

The trip input module collects preferences; the AI module generates the itinerary; the persistence module stores and retrieves saved trips; the place-search module constructs Google Maps URLs; and the print module uses the browser print interface. The modules are related through the main trip-planning workflow.

6. Testing and Evaluation

The main evaluation should include successful itinerary generation, validation errors, saved-trip persistence, retrieval, deletion, map searches and printing. Performance results should be measured rather than assumed. The current documentation does not claim a completed benchmark or user study.

7. Limitations

AI-generated recommendations and costs are estimates. The Google Maps integration returns search URLs and does not directly verify listings, prices, ratings, opening hours, availability or dietary suitability. Booking, payment and live provider inventory are not currently implemented.

8. Future Scope

Potential improvements include automated tests, user authentication, user-owned trip records, approved live place data, weather context, authorized booking integrations, improved price estimation, multilingual support, accessibility and operational monitoring.

9. Conclusion

Trip Genius AI demonstrates an end-to-end AI-assisted itinerary workflow with persistent saved trips and external place-search links. Additional testing and evidence capture are needed before making production-level reliability claims.